OAK HICKORY REGENERATION - SANDER

September 22, 2020

The Sander metric considers only saplings in the DBH categories 0–1 cm, 1–2.5 cm, and 2.5–5 cm. Saplings were counted in eight 10-m^2 boxes for each surveyed plot. If the number of oak-hickory saplings exceeds ~1000 per hectare, the plot has adequate advanced regeneration of oak-hickory. This threshold value is equivalent to 1 oak-hickory sapling per 10-m^2 box. Table 1 presents bootstrapped estimates of oak-hickory sapling densities, with 90% confidence intervals. For simplicity, the densities are presented as number of oak-hickory saplings per 10-m^2 box. A value greater than 1 means the Sander et al. (1976) threshold is surpassed.

Table 1. Oak-hickory sapling densities in oak-hickory communities, by SubPark *Estimates and 90% confidence intervals are determined from hierarchical bootstrapping of the survey data. 'Nplots' is the number of oak-hickory plots surveyed in that SubPark in Cycle 1. The last 3 columns show the 5%, 50%, and 95% quantiles for mean density (per 10-m^2 box) of oak-hickory saplings in Oak-Hickory communities of that SubPark. For example, 12 oak-hickory plots were surveyed in CARL oak-Hickory communities. Ninety percent of the bootstrapped estimates of mean oak-hickory sapling densities were between 0.15 and 0.65 oak-hickory saplings per 10-m^2 box. Half of the bootstrapped estimates were below 0.35 oak-hickory saplings per 10-m^2 box (and half were above this value). A density greater than 1 means it exceeds the Sander et al. (1976) threshold for adequate advanced regeneration of oak-hickory. Therefore, even when sampling variability is taken into account, CARL oak-hickory plots seem to have, on average, oak-hickory sapling densities well below the regeneration threshold. NOTE 1: Only showing data for completed survey cycles. NOTE 2: For each plot in a bootstrap sample, if QAQC data were also available, those counts were equally likely to be sampled from because we assume the QAQC counts are just as likely as the non-QAQC counts to be correct. NOTE 3: Bootstrapping confidence intervals (or any other method of estimation) isn't recommended for very small sample sizes (e.g., few surveyed plots in a SubPark) but there are not clear rules of thumb for minimum sample sizes. Be wary of estimates for SubParks with few surveyed plots.*

CGtype	SubPark	Nplots	5%	50%	95%
OakHick	CHIC	10	0.00	0.02	0.06
OakHick	KIMO	20	0.00	0.04	0.16
OakHick	DONE	11	0.00	0.05	0.10
OakHick	GUCO	9	0.00	0.06	0.17
OakHick	COWP	3	0.00	0.08	0.25
OakHick	KNOB	9	0.03	0.08	0.15
OakHick	CUGA	36	0.04	0.09	0.16
OakHick	RUCA	20	0.03	0.16	0.35
OakHick	STRI	6	0.00	0.21	0.58
OakHick	LOOK	14	0.11	0.22	0.37
OakHick	MACA	29	0.09	0.22	0.40
OakHick	CARL	19	0.10	0.24	0.43
OakHick	HEIM	9	0.11	0.25	0.46
OakHick	LIRI	27	0.25	0.41	0.65
OakHick	SHIL	19	0.34	0.50	0.68

Figure 1. Oak-hickory sapling densities in oak-hickory communities, by SubPark *Median and 90% bootstrapped confidence intervals for estimated mean density of oak-hickory saplings. These data correspond with Table 1. The dashed blue line is the threshold density for adequate advanced regeneration.*

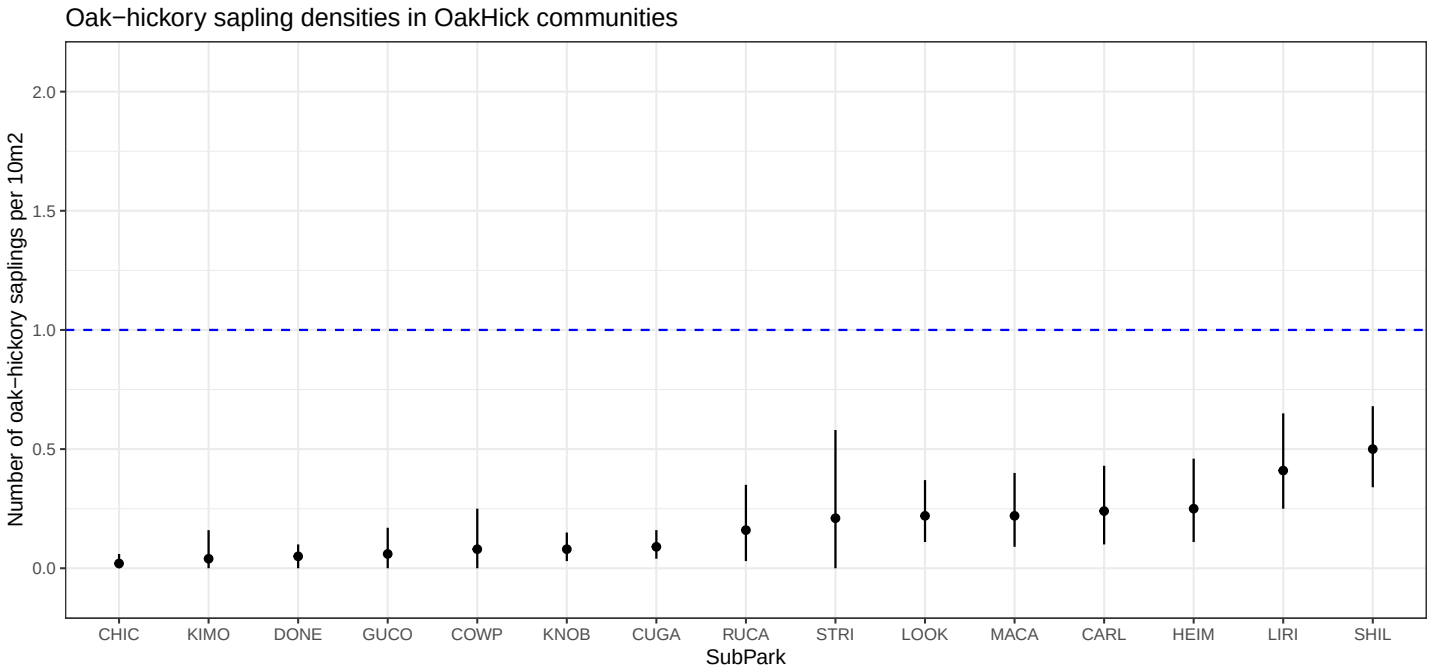


Table 2. Oak-hickory sapling densities in ruderal communities, by SubPark

CGtype	SubPark	Nplots	5%	50%	95%
Ruderal	BIRT	7	0.00	0.00	0.00
Ruderal	SHIL	2	0.00	0.00	0.00
Ruderal	STRI	13	0.00	0.01	0.05
Ruderal	LOOK	4	0.00	0.03	0.12
Ruderal	DONE	4	0.00	0.09	0.19
Ruderal	GUCO	10	0.02	0.09	0.21
Ruderal	NISI	17	0.03	0.10	0.18
Ruderal	KNOB	4	0.00	0.25	0.75
Ruderal	KIMO	3	0.04	0.29	0.67
Ruderal	CARL	1	0.00	0.50	1.00
Ruderal	MACA	19	0.29	0.51	0.80
Ruderal	CHIC	7	0.11	0.66	1.62
Ruderal	COWP	13	0.21	0.74	1.62
Ruderal	LIRI	2	0.25	1.00	1.94

Figure 2. Oak-hickory sapling densities in ruderal communities, by SubPark

